bioREACT

Deciding what is worth making, and whether it can actually be made

An agentic research workbench spanning molecular feasibility and bioreactor process design.

The problem

Two decisions sit between a target and a medicine, and both are made on incomplete evidence. Docking ranks how tightly a molecule binds but says nothing about whether anyone can make it, so generative pipelines optimise the score and their best output is often unbuildable — discovered only when a process chemist reads the list weeks later. Separately, a biochemical engineer must choose a bioreactor and its operating conditions, balancing oxygen supply, mixing, shear and scale-up, usually by running expensive experiments to find out.

One pattern, applied twice

A generative proposer produces high-scoring candidates; deterministic critics reject the ones that cannot be made, and say exactly why. Rejections are structured — criterion, value, threshold, suggested fix — and the proposer must address each by name in the next round. In both of our domains the best-scoring proposal is unbuildable, and the critics are what make the search useful.

Track 1 — Molecular feasibility (MRP1)

A designer agent proposes molecules and chooses which to dock; synthesisability and process-chemistry critics judge whether each can be made at all, and at scale. In our own data the two objectives are anti-correlated: the three best-docking molecules (−12.90, −12.32, −11.20 kcal/mol) each need 15–18 synthetic steps, and the critics reject all three. Two redesign rounds took candidates from 0/3 to 3/3 passing, converging on an achiral biphenyl sulfonamide at **−10.82 kcal/mol in three steps with zero stereocentres**, against reference inhibitor MK-571 at −9.10. MRP1 drives multidrug resistance in cancer; inhibiting it is meant to restore chemotherapies that have stopped working.

Track 2 — Bioreactor feasibility (trastuzumab)

For a biologic, bioREACT derives the process physics rather than scoring it with tunable weights. Oxygen transfer comes from the Van't Riet kLa correlation, power and tip speed from impeller geometry, mixing time from Nienow. Cell density is not a free input: it is set by the oxygen the vessel can actually deliver at a 30% dissolved-oxygen floor, with perfusion following at constant cell-specific perfusion rate, as these processes are run industrially. Nine named constraints decide feasibility — shear, flooding, foaming, CO_2 stripping, mixing, membrane pressure, viability, lactate and glucose. A constrained Bayesian optimiser then searches the agitation–aeration plane using a continuous constraint-margin surrogate. At 500 L it finds a feasible operating point in **10/10 random seeds where matched-budget random search fails in 4/10**.

Evaluation

The chemistry critics are graded on 34 molecules whose answer we already know, against a stated question: could a medicinal chemistry team make this de novo, at gram scale, in a short campaign? They score **30/34 (88%)** — 23/23 on an easy tier, 7/11 on a deliberately hard one, with zero false passes at the strict setting. We say plainly that the easy tier is unimpressive, since those groups separate on molecular weight alone, and we name all four misses. On the engineering side the model reproduces a known scale-up squeeze: the feasible operating window shrinks from 7.6% of the plane at 3 L to zero at 2000 L, because at geometric similarity a constant P/V raises tip speed with vessel diameter.

What it cannot do

Docking ranks, it does not predict affinity: rigid receptor, one grid, no rescoring. Repeat Vina runs varied by 0.17 kcal/mol on a rigid candidate and 1.51 on the flexible reference, so our margin over MK-571 is close to our own measured noise — the defensible claim is comparable affinity at a fraction of the synthetic cost. Step counts are a heuristic from rings, stereocentres and heavy atoms, not a retrosynthesis. The bioprocess model is synthetic and unvalidated, and flags the kLa correlation wherever it is extrapolated beyond its fitted gas-velocity range. We do not cost molecules. MRP1 and trastuzumab are deliberately separate demonstrations: the manufacturing gate refuses to route a small molecule into a CHO bioreactor, because mammalian cells cannot express one. That refusal is designed behaviour, not a gap.

Human oversight

Five gates — candidate, reactor, optimiser, protocol and report — each require a person to approve, request changes or reject. Humans set the rubric thresholds and write the ground-truth labels. Hard caps on rounds, tool calls and docking runs are shown in the interface, so the agent cannot run away. Every value is tagged measured, derived, researcher-supplied or assumed. The tool ranks and shows its evidence; the decision stays with the researcher.